

1. **Purpose:** This document defines the fasteners and standard hardware used in the mechanical components of the Amazon Robotics system.
2. **Scope:** The specification to define the part number, dimensions, material and finish of common fasteners and hardware are defined in this document.
3. **Part Number:** The part number is linked to the characteristics of the fastener and in conjunction with this specification is all that is required to procure the part.

The part name follows this convention where each section of the name has an abbreviation for

- The part name follows this convention where each section of the name has an abbreviation for description: [Category][Head][Size]X[Length][Material][Grade][Finish][Other]. The definition for the abbreviation used in the part number is as follows in the subsequent sections. There shall be no spaces between characters in the part number. All characters are capitalized.

Example part names:

3.1. Category: This is the broad category for the type of hardware. The abbreviation is used in the part name and the description can be referenced in this document. The abbreviation is 3 numbers and a dash.

DESCRIPTION:	ABBREV.
SCREW	201-
WASHER	202-
NUT	203-

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DESCRIPTION:	ABBREV.
RETAINING RING	204-
PINS	205-

3.2. Head: This is the style of the head and drive of the hardware. The abbreviation is used in the part name and the description can be referenced in this document. Abbreviation is 3 letters and is associated with the following standard.

Table 2: STYLE OF HARDWARE

DESCRIPTION:	ABBREV.	STANDARD	EQUIVALENT FOR REFERENCE
Screws			
SOCKET HEAD CAP (SCREW)	SHC	ASME B18.3(E), ISO 4762 (M)	
BUTTON SOCKET HEAD CAP (SCREW)	BSC	ASME B18.3(E), ISO 7380 (M)	
FLAT SOCKET HEAD CAP (SCREW)	FSC	ASME B18.3(E), DIN 7991(M), ISO 10642(M)	
SOCKET HEAD SHOULDER (SCREW)	SHS	ASME B18.3(E), ISO 7379(M)	
PAN HEAD PHILIPS (SCREW)	PHP	ASME B18.6.3(E), DIN 7985(M)	
FLAT HEAD PHILIPS 82 (SCREW)	FHP	ASME B18.6.3(E), DIN 965(M)	
HEX HEAD(SCREW)	HXH	ASME B18.2.1(E), DIN 931(M), DIN 933(M)	
HEX FLANGED HEAD (SCREW)	HFH	ASME B18.2.1(E), DIN 6921(M)	
PAN HEAD PHILLIPS SHEETMETAL (SCREW)	PPS	ASME B18.6.3(E)	
THUMB (SCREW)	TMB	ASME B18.6.8(E)	
HEX HEAD SELF DRILLING (SCREW)	HSD	ASME B18.6.3(E)	
FLAT HEAD PHILIPS SELF DRILL (SCREW)	FSD	ASME B18.6.3(E)	
Nuts			
FLANGE HEAD (NUT)	FLH	ASME B18.16.4(E), DIN 6923(M)	
ACORN LOW HEIGHT (NUT)	ALH	DIN 1587(M)	
LOW PROFILE/JAM (NUT)	JAM	ASME B18.2.2(E), DIN 439B(M)	
PLAIN HEX (NUT)	HEX	ASME B18.2.2, DIN 934	
NYLOC (NUT)	NNY	ASME B18.16.3M, DIN 985	
Washers			
PLAIN (WASHER)	PLN	ASME B18.21.1(E), DIN 125(M)	

DESCRIPTION:	ABBREV.	STANDARD	EQUIVALENT FOR REFERENCE
SPLIT (WASHER)	SPL	ASME B18.21.1(E), DIN 127(M)	
BELLEVILLE (WASHER)	BEL	ISO/CD 19690-1/2	DIN 2093
INTERNAL TOOTH (WASHER)	INT	ASME B18.21.1 (E) ,, DIN 6797(M)	
EXTERNAL TOOTH (WASHER)	EXT	ASME B18.21.1 (E), DIN 6797(M)	
Retaining Rings			
BASIC INTERNAL (RETAINING RING)	RRI	ASME B18.27b(E), DIN 472(M)	
BASIC EXTERNAL (RETAINING RING)	RRE	ASME B18.27b(E), DIN 471(M)	
Pins			
SPRING (PIN)	SPR	ASME B18.8.2(E), DIN 1481(M)	
STRAIGHT (PIN)	STR	ASME B18.8.2 (E), DIN 6325(M)	

3.3. Size: This is the nominal thread size and pitch of the fastener. The dimensions shall meet the requirements of the industry standard listed in the table below.

3.3.1.Screws, Nuts and Bolts: This is the nominal thread size and pitch of the fastener.

English: For inch based fasteners, the thread is defined as the *nominal size – threads per inch*. Fasteners less than 1/4" in diameter are listed as the size number; fasteners 1/4" in diameter or greater are listed as the decimal equivalent of the size rounded to 2 decimals; e.g. 10-32, .25-20 and .31-18. Numbered fasteners are listed as the numerical size without trailing decimals; e.g. #2 would be 2.

Additional Examples:

#4-40 → 4-40

#10-32 → 10-32

1/4"-20 → .25-20

1/2"-13 → .50-13

Thread class is a minimum of 2A or 2B for all machine threads.

Metric: For Metric based fasteners the pitch is the coarse thread associated with the nominal diameter in millimeters prefaced with the letter *M*.; i.e., M6.

Thread class is a minimum of 6H/6G for all machine threads.

Table 3: THREAD STANDARDS

UNIT	EXAMPLE ABBREVIATION	STANDARD
METRIC	M6	ANSI/ASME B1.7-2006
ENGLISH	.25-20	ANSI/ASME B1.13M-1995

3.3.2. Pins: This is the nominal diameter of the pin.

3.3.3. Retaining Rings: This is the nominal shaft diameter related to the size of the retaining ring.

3.3.4. Washers: This is the nominal size based on the diameter of the fastener it is related to. Washers for numbered fasteners are listed as the numerical size without trailing decimals; i.e. #2 would be 2, not .094; 2" ID would be 2.00.

3.4. Length: This is the nominal length in inches with 2 decimals for English and Millimeters for metric. The length shall be prefixed with the character X. For washers (202-) characters for length are not applicable and therefore shall be omitted.

3.5. Material: This a general statement about the material. The composition of the material used shall be determined by the requirements of the Grade. Abbreviation is 2 letters.

Table 4: MATERIAL OPTIONS

DESC.	ABBREV.
ALLOY STEEL	AS
STAINLESS STEEL	SS

3.6. Grade: The grade of the fastener shall meet the requirements of the following standards. The material specified above will affect the standard to be used below. The abbreviation is up to 4 characters.

3.6.1. Metric Stainless Steel: This specification is for the property group, grade and class per the industry standard ASTM F738M. The group is the steel composition, the grade is the alloy group within that composition and the class is the minimum tensile strength designation. Any material within this specification may be used.

GROUP	GRADE	CLASS	ABBREVIATION	STANDARD	EXAMPLE STEEL SERIES
A	2	70	A27	ISO 3506	A2-7 (similar to 304)

3.6.2. Metric Alloy Steel: For metric components with Alloy Steel material, this is the control for material strength and material composition per the industry standard listed below.

PROPERTY CLASS	ABREV.	STANDARD	COMMENTS
4.8 OR WITHIN RANGE ALLOWED PER THE INDUSTRY STANDARD	STD	ISO 898-1	MINIMUM STRENGTH
8.8	8.8		METRIC HIGH STRENGTH
10.9	10.9		METRIC HIGH STRENGTH
12.9	12.9		METRIC EXTRA HIGH STRENGTH (NOT COMMON)
10	10	ISO 898-2	METRIC HIGH STRENGTH NUTS
12	12		METRIC EXTRA HIGH STRENGTH NUTS (NOT COMMON)

3.6.3.English Stainless Steel: For English sized components with Stainless Steel material, this is the control for material strength and material composition per the industry standard listed below. A series is a range of acceptable material, for instance 300 series includes specific grades that start with 3 i.e.: 304 or 316.

PROPERTY CLASS	ABREV.	STANDARD	EXAMPLE STEEL
300 SERIES	300	ASTM F593	(18-8) 304,316
410	410		410 ONLY

3.6.4.English Alloy Steel: For English sized components with Alloy Steel material, this is the control for material strength per the industry standard listed below.

GRADE	ABBREV.	STANDARD	COMMENTS
5 OR WITHIN RANGE ALLOWED PER THE INDUSTRY STANDARD	STD	SAE J429	FOR NON STRUCTURAL COMPONENTS
8	8		HIGH STRENGTH STEEL

3.7. Finish: The following descriptions represent acceptable options for fastener finish. Abbreviation is 1 letter with the exception of zinc mechanical flake.

Table 5: FINISH OPTIONS

DESCRIPTION.	ABBREV.	COMMENTS
BLACK OXIDE	B	
ZINC ELECTROPLATE	Z	ONLY USE WITH SCREWS BELOW CLASS 10 STRENGTH
ZINC MECHANICAL FLAKE	ZM	ONLY USE WITH SCREWS EQUAL TO OR ABOVE CLASS 10 STRENGTH
PLAIN	L	
PASSIVATED	P	USED WITH MOST SS COMPONENTS

3.8. Other: This can be used for custom properties of a fastener. A value in this field of the name is optional. Abbreviation is 2 numbers.

Table 6: OTHER CATEGORY OPTIONS

DESC.	ABBREV.	ADDITIONAL DETAILS	APPLIES TO
SELF LOCKING	01	NYLON PATCH, 90° TO 180° RADIAL COVERAGE, BEGINNING 1 TO 3 THREADS FROM END AND HAVING LENGTH OF 4 TO 6 THREADS LONG, MEET IFI 124	SCREWS
SELF LOCKING SEMS - EXT	02	INTEGRATED SEMS EXTERNAL TOOTH WASHER	SCREWS
SELF LOCKING LOCTITE	03	LOCTITE THREAD LOCKING PATCH, 360° RADIAL COVERAGE, BEGINNING 1 TO 3 THREADS FROM END AND HAVING LENGTH OF ≥4 THREADS LONG, MEET IFI 125	SCREWS
FULLY THREADED	04	FOR SCREWS OF LONGER LENGTH THAT WOULD HAVE PARTIAL THREAD PER THE SCREW STANDARD	SCREWS
SELF LOCKING DIMPLE	05	DEFORMED THREAD	NUTS
FULLY THREADED + NYLOC	06	NYLOC THREAD LOCKING PATCH FOR SCREWS OF LONGER LENGTH THAT WOULD HAVE PARTIAL THREAD PER THE SCREW STANDARD. NYLOC PATCH APPLIED AS DESCRIBED IN 01	SCREWS
FULLY THREADED + LOCTITE	07	LOCTITE PATCH FOR SCREWS OF LONGER LENGTH THAT WOULD HAVE PARTIAL THREAD PER THE SCREW STANDARD. LOCTITE PATCH APPLIED AS DESCRIBED IN 03	SCREWS
SELF LOCKING FLEX-TOP	08	FLEXLOC SPLIT TOP	NUTS
SELF LOCKING SEMS - INT	09	INTEGRATED SEMS INTERNAL TOOTH WASHER	SCREWS

- 4. Regulatory Compliance:** All components shall meet Amazon Robotics global product governance and compliance standard QMS 960-00165.
- 5. Description:** The description of the part shall match the part number with the exception of the category where the first 10 letters of the description in Section 3.1 are used in place of the abbreviation. For example: Part number “201-SHSM6X10AS12.9B01”'s description is “SCREW-SHSM6X10AS12.9B01”

Appendix:

Example of part naming options:

	SHC							
	BSC							
	FSC							
	SHS							
	PHP							
	FHP							
	HXH							
	HFH							
	HEX							
	NNY							
	PPS							
	TMB							
	HSD							
	FLH							
	ALH							
	PLN					12		
	SPL					10		08
	INT					12.9		07
	EXT					10.9		06
205-	RRI	M#	X			STD	ZM	05
204-	RRE	M##	X	###		8	B	04
203-	SPR	.##-##	X	.##		300	Z	03
202-	STR	##-##	X	##	AS	A27	L	02
201-	BEL	###-##	X	#	SS	410	P	01
Category	Head	Size		Length	Material	Grade	Finish	Other

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REVISION HISTORY:

Rev.	ECO #	Description of Change	Sections Affected	Date	Originator/Department
01	008631	INITIAL RELEASE	All	2016-02-08	C.NAUSEDA/ HW ENG.
02	009218	3.2 Replace ANSI with ASTM standards, 3.6 Change metric SS abbreviation to A27 from 304, 3.8 add details to 01	3	2016-05-05	C.NAUSEDA/ HW ENG.
03	009454	Add plain HEX nut and nyloc NNY nut, 3.8 add details to 01 (update nylon patch typo), add 09 (internal SEMS), reword 02 and make external SEMS only, add additional details to 01, 02, 03, 07, improve formatting, added typical use cases to Table 6	3.2, 3.8	2016-06-28	M. BRADY/ HW ENG.
04	009631	Added clarity to number sized bolts/screws, and washers. Add FSD head type.	3.3, 3.2	2016-10-27	C.NAUSEDA/ HW ENG.
05	017780	Adjusted zinc finish suffixes so that they are more easily identifiable between electroplate and mechanical flake in cases where hardware class strength is important. Updated the metric standard for socket cap head screws from DIN 912 to ISO 4762. Addition of class 8.8 to strength table.	3.2, 3.6.2	2021-05-24	M. TAVERAS / HW ENG.